

Social coding and open software - What can you do to get credit for your code and to allow reuse

In this lesson we will discuss how and why to share code and data, what kind of licenses are used in what situation, and how software can be cited.

We will try to connect software licenses to FAIR principles, give practical recommendations for starting, contributing, and reusing code and help you navigating and deciding on licenses.

Why software licenses matter

- You **find some great code** or data that you want to reuse for your own publication (good for the original author: you will cite them and maybe other people who cite you will cite them).
- You need to **modify the code** a little bit, or you remix the data a bit.
- When it comes time to publish, you realize there is no **license**.

Now we have a problem:

- You manage to **publish the paper without the software/data** but others cannot build on your software and data and you don't get as many citations as you could.
- Or, you **cannot publish it at all** if the journal requires that papers should come with data and software so that they are reproducible.

This lesson is about how to avoid this situation for you and others.

Prerequisites

No computational prerequisites required for this lesson.

20 min	Social coding
45 min	Software licensing focusing on open source
20 min	Software citation
10 min	Sharing data

Social coding

Objectives

- Get an overview of motivations, benefits, but also risks of sharing and reusing code.
- Learn to identify whether you can use other peoples software.

Discussion/questions/poll: basics of sharing

Instructor note

These questions 1-4 below can be used as a starting point and copied to the collaborative document or form input for an online poll.

Alternatively, they can be discussed in voice or free-form text (discussion box below).

Social-1: Think about if and how you share

- Did you ever share your code? If yes, what motivated you? Come up with **reasons for sharing** your scripts/code/data.
- Also think about **reasons for not sharing**.

Question 1: Why would I want to share my scripts/code/data?

****Choose many****. Vote by adding an `o` character:

- A: Easier to find and reproduce (scientific reproducibility)
 - votes:
- B: More trustworthy: others can verify correctness and find and report bugs
 - votes:
- C: Enables others to build on top of your code
(derivative work, provided the license allows it)
 - votes:
- D: Others can submit features/improvements
 - votes:
- E: Others can help fixing bugs
 - votes:
- F: Many tools and apps are free for open source, so no financial cost for this
(GitHub, GitLab, Appveyor, Read the Docs)
 - votes:
- G: Good for your CV: you can show what you have built
 - votes:
- H: Discourages competitors. If others can't build on your work,
they will make competing work
 - votes:
- I: When publicly shared, usually we time-stamp or set a version,
so it is easier to refer to a specific version
 - votes:
- J: You can reuse your own code later after change of job or affiliation
 - votes:
- K: It encourages me to code properly from the start
 - votes:

Question 2: The most concerning thing for me, If I share my software now

****Choose one****. Vote by adding an `o` character:

- A: It will be scooped (stolen) by someone else
 - votes:
- B: It will expose my "ugly code"
 - votes:
- C: Others may find bugs and mistakes. What if the algorithm is wrong?
 - votes:
- D: I will get too many questions, I do not have time for that
 - votes:
- E: Losing control over the direction of the project
 - votes:
- F: Low quality copies will appear
 - votes:

- G: I won't be able to sell this later. Someone else will make money from it
 - votes:
- H: It is too early, I am just prototyping, I will write version to distribute later
 - votes:
- I: Worried about licensing and legal matters, as they are very complicated
 - votes:

Question 3: Why is software often treated differently from papers?

Free-form answers:

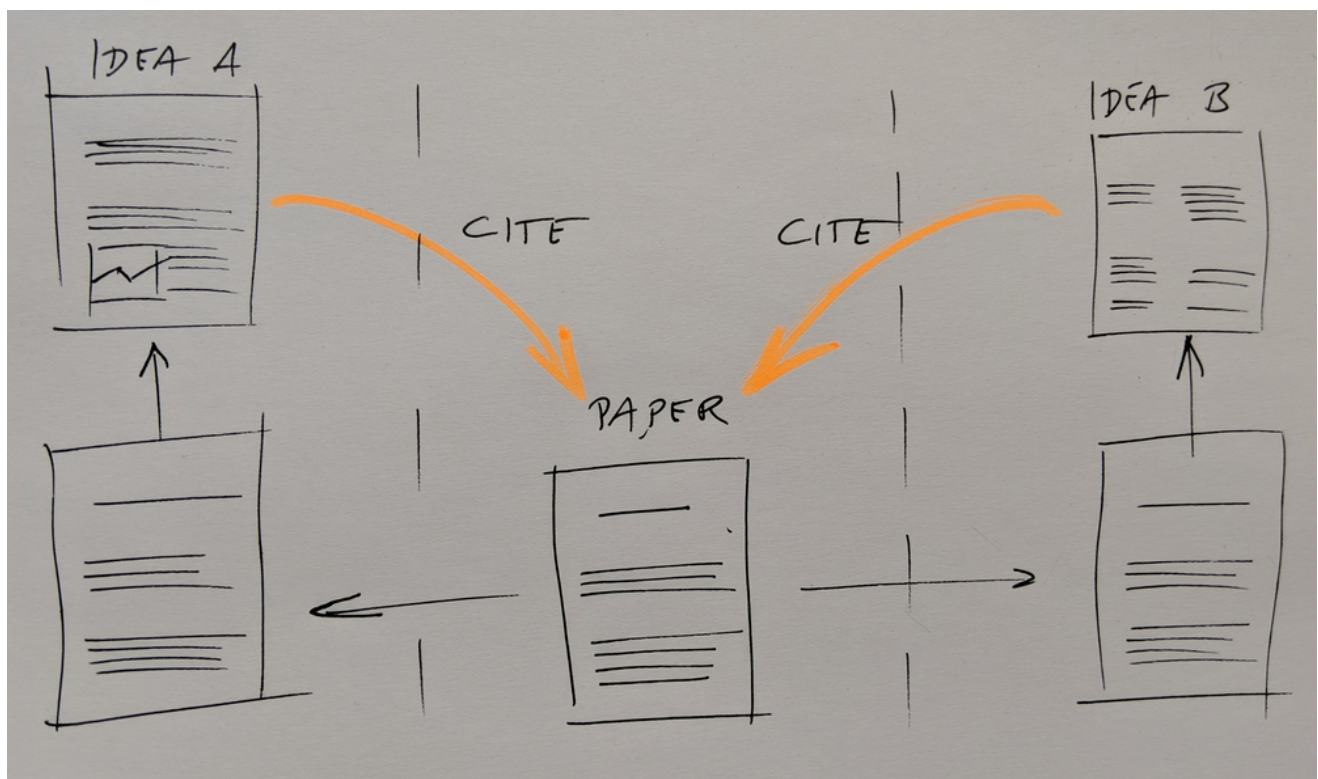
- ...
- ...
- ...
- ...
- ...
- ...

Question 4: When you find a repository with code/library you would like to reuse, what are the things you look at to decide whether you use it?

Free-form answers:

- ...
- ...
- ...
- ...
- ...
- ...

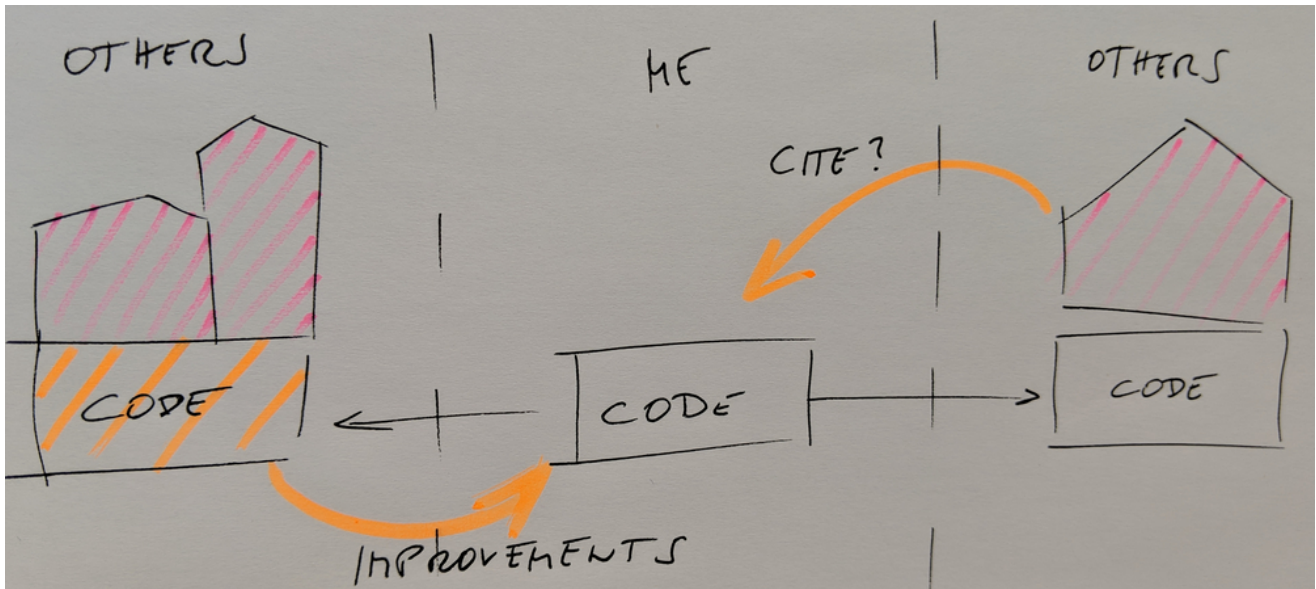
Comparing sharing papers and sharing code



Citation as one form of academic credit to motivate sharing papers.

Sharing papers and academic credit:

- The goal is maximum visibility and maximum reuse.
- The more interesting science is done referencing my paper, the better for me.
- Nobody actively tries to limit the reach of their papers.



Different ways we can benefit from sharing code.

Sharing code:

- “I did all the ground work and they get to do the interesting science?”
- Sharing code and encouraging *derivative* work may boost your academic impact.
- But will your work be visible if it is used two levels deep down?

Journal policies as motivation for sharing

From [Science editorial policy](#) [↗]:

“We require that **all computer code used for modeling and/or data analysis** that is not commercially available be deposited in a **publicly accessible repository** upon publication. In rare exceptional cases where security concerns or competing commercial interests pose a conflict, code-sharing arrangements that still facilitate reproduction of the work should be discussed with your Editor no later than the revision stage.”

From [Nature editorial policy](#) [↗]:

“An inherent principle of publication is that others should be able to replicate and build upon the authors’ published claims. A condition of publication in a Nature Research journal is that authors are required to make **materials, data, code, and associated protocols promptly available** to readers without undue qualifications. Any restrictions on the availability of materials or information must be disclosed to the editors at the time of submission. Any restrictions must also be disclosed in the submitted manuscript.”

However [a study](#) [↗] showed that despite these policies, many people still do not share their code 😞. This paper includes samples of charming author responses such as:

“When you approach a PI for the source codes and raw data, you better explain who you are, whom you work for, why you need the data and what you are going to do with it.”

Motivation for open source software

Instructor note

We revisit answers to Question 1 (above).

- Enable derivative work
- Do not lock yourself out of own code
- Attract developers who want to be able to show the coding work on their CVs
- Tightly regulated domains require open source
- Open-source software (OSS) can lead to more engagement from industry which may lead to more impact
- If it's not open, it is not likely to become standard

Sharing software is also scary

Instructor note

We revisit answers to Question 2 (above).

- **Fear of being scooped:** A license can avoid it, and you can release when you are ready. Anyway, it is very unlikely that others will understand your code and publish before you without involving you in a collaboration. Sharing is a form of publishing.
- **Exposes possibly “ugly code”:** In practice almost nobody will judge the quality of your code. “Software, once written, is never really finished” (N. Asparouhova).
- **Others may find bugs and mistakes:** Isn't this good? Would you not like to use a code which gives people the chance to locate bugs? If you don't release, people will assume there are bugs anyway.
- **Others may require support and ask too many questions:** This can become a problem: use tools and community and protect your time. You aren't required to support anyone. You can also “archive” a repository to disable most forms of interaction (issues, PRs). Also a note in README on support level helps.
- **Fear of losing control over the direction of the project:** Open source does not mean everybody can change **your version**.
- **“Bad” derivative projects may appear:** It will be clear which is the official version.

Code reusability: What contributes to reusability?

What contributes to you being able to reuse stuff that others make, and others (or you) being able to reuse your stuff? When you find a repository with code you would like to reuse, you may look at the following things to determine its reusability:

Instructor note

This can be now reconnected to Question 4 (above).

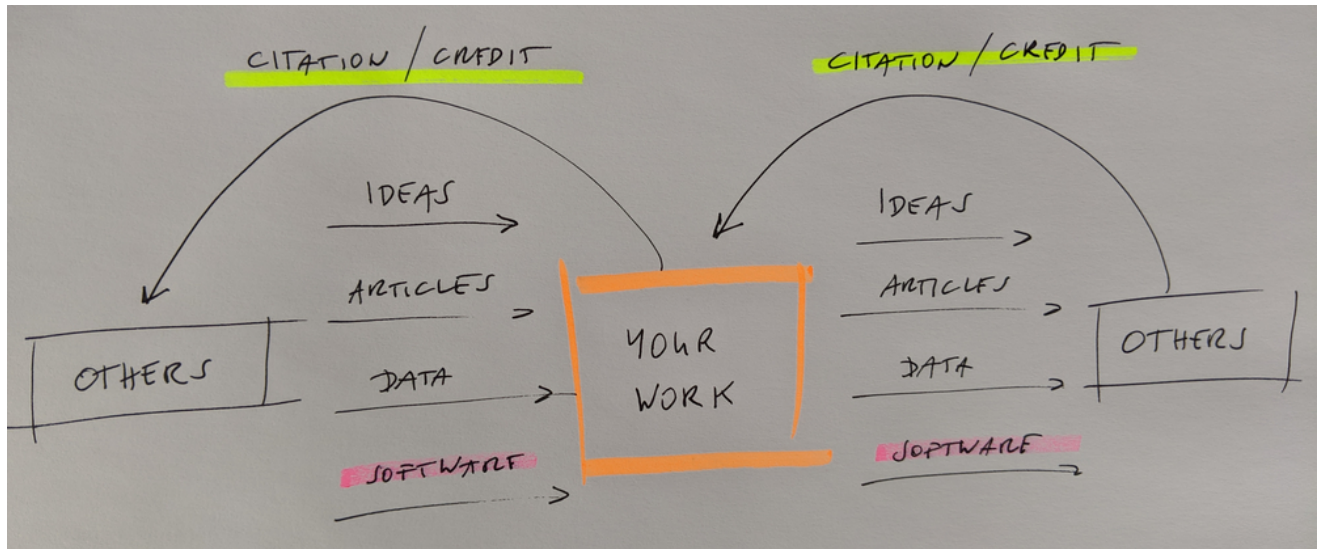
- **Date of last code change:** Is the project abandoned?
- **Release history:** How about stability and backwards-compatibility?
- **Versioning:** Will it be painful to upgrade?
- **Number of open pull requests and issues:** Are they followed-up?
- **Installation instructions:** Will it be difficult to get it running?
- **Example:** Will it be difficult to get started?
- **License:** Am I allowed to use it?
- **Contribution guide:** How to contribute and decision process?
- **Code of conduct:** How to make clear which behaviors are unacceptable and discouraged?
How violations of Code of conduct will be handled?
- **Trust and community:** Somebody you trust recommended it?

... most of which you have learned or will learn during this [CodeRefinery](#) workshop!

Social-2: Discussion about “You aren’t required to support anyone”

- Have you experienced an implicit expectation of support?
- Supporting all requests can lead to overworking and mental health issues.
- Not supporting requests can also induce guilt.
- Most projects are maintained by 1 or 2 persons.
- Most projects cannot retain contributors for a longer time. Interests change. “Casual contributors are like tourists visiting NYC for a weekend” (Nadia Asparouhova, book below).
- If you maintain all projects that you start forever, at some point it may be difficult to start new projects.
- What are your experiences? Do you agree with the above thoughts?
- Book recommendation: Nadia Asparouhova (formerly Nadia Eghbal): “Working in Public: The Making and Maintenance of Open Source Software (Stripe Press)”

How our work connects to the work of others



Whether and what we can share depends on how we obtained the components.

- Our work depends on outputs from others. Research of others depends on our outputs.
- Whether you can share your output depends on how you obtained your input.
- A repository that is private today might become public one day.
- Sometimes “OTHERS” are you yourself in the future in a different group/job.
- Our code often starts by changing other code: **derivative work**.
- **Software licenses** matter. And this is what we will discuss in the next episode.

Software licensing focusing on open source

! Objectives

- Principles of open source licensing
- Difference between permissive and copyleft licenses
- Regulations for AI-generated and AI-assisted code
- Determine the software license for your project following EU regulation
- Navigate the Joinup Licensing Assistant to select a compliant license
- Understand the licensing distinction between container recipes and container images

💬 Limitations and context of this lesson

This lesson is designed as practical educational material for researchers and research software engineers, **not formal legal advice**

- EU directives set only minimum requirements in some areas: Member States implement them differently and may add national rules not covered here. For example, some Member States let university researchers retain ownership of the programs they write instead of applying the employer rule in Art. 2(3).
- Institutional Context: Employment contracts, grant agreements, and university policies heavily influence software ownership and licensing choices.

- This lesson covers only the general principles of open-source reuse, copyright scope, and software adaptation.

If you need formal guidance references below and legal experts, especially if you have legal services at your host institute, could be of help:

- [EUR Directive 2009/24/EC](#) 
- [Compendium of U.S. Copyright Office Practices \(3rd Ed.\) – Chapter 700, Section 721: Computer Programs](#) 
- [Chinese Regulations on Computer Software Protection,\(search:"????????"\)](#) 
- [Joinup Licensing Assistant,JLA](#) 
- [FSFE REUSE Initiative](#) 
- [Research Software Alliance Policy Directory](#) 

Introduction: What is a Software License?

Under copyright law worldwide, software without an explicit license defaults to All Rights Reserved: nobody else may run, copy, modify, distribute, or build on your code. A software license is how the copyright holder exercises their exclusive rights, granting others permission to reproduce, distribute, modify, and sometimes sublicense the work.

Note that author and copyright holder may differ: under Art. 2(3), an employer exercises the economic rights in code written by an employee on the job, unless a contract says otherwise. The employee is still the author; the employer is who licenses it. This matters in practice, because the person choosing the license for a research project is often not the person who wrote the code.

In this lesson, we focus on open-source licenses to define both how we grant permissions for software we develop (outbound licensing) and how we safely comply with terms attached to code written by others (inbound reuse).

Open-source licenses fall into three main families:

- **Permissive (e.g., MIT, Apache-2.0, 0BSD):** *Do whatever you want, just keep credit.* Grants maximum reuse freedom, allowing anyone to modify, embed, or re-license your code in open or closed projects.
- **Copyleft / Reciprocal (e.g., GPL-3.0, EUPL-1.2):** *Share alike.* Grants full freedom to run and modify, but requires that any distributed adaptation or combined work also be released under matching copyleft terms.
- **Weak copyleft (e.g., LGPL-3.0, MPL-2.0, EPL-2.0):** *Share alike, but only within a boundary.* Reciprocity applies to the file (MPL-2.0) or the library (LGPL), not to your whole project. Your surrounding code can usually stay permissive or even closed, while modifications to the covered files or library must stay open.

You will hear copyleft called *viral* or *infectious* in developer conversation. The slang is worth knowing, but it is misleading in two ways: nothing spreads by mere contact, so code merely sitting beside GPL code in a repository or a container image is unaffected, and the requirement only triggers when you **distribute**, not when you run modified code internally. Reciprocity reaches only across specific technical boundaries such as embedding snippets or static linking, and how far it reaches depends on which copyleft license you are dealing with. Choosing copyleft over permissive is a project-level decision, not a sign that a license is harmful.

Weak copyleft is worth a closer look, because it is widely used and its terms are more conditional than the label suggests. LGPL-3.0 §4 lets you ship a combined work under your own terms only if those terms do not restrict modification of the LGPL portions, or reverse engineering for debugging those modifications, and this condition applies whether you linked statically or dynamically. Since most proprietary end-user licenses forbid reverse engineering, the common shorthand that “dynamic linking is safe” is not the whole story. The practical lesson is that “does this dependency force my project open?” has no general answer: it depends on which copyleft license, and at which boundary.

The diagram below unifies these license choices and their downstream rights:

```
%%{init: {'themeVariables': { 'edgeLabelBackground': '#faf5ff' }}}%% flowchart TB
    subgraph box ["How License Selection Governs Code Reuse"]
        A["<b>Your Research Codebase</b><br><i>(Source code, container recipes, prompt templates)</i>"] --> B["<b>All Rights Reserved</b><br>❌ Zero permissions: Cannot run, modify, or share"]
        A --> C["Attach License<br>(Explicit Permission Grant)"]
        C --> D["<b>Permissive</b><br><i>MIT, Apache-2.0, BSD</i>"]
        C --> E["<b>Weak Copyleft</b><br><i>LGPL-3.0, MPL-2.0, EPL-2.0</i>"]
        C --> F["<b>Copyleft</b><br><i>GPL-3.0, EUPL-1.2</i>"]
        C --> G["<b>Closed Source / Restricted</b><br>🚫 <i>Not discussed in this lesson</i>"]
        D --> D1["Run & Modify? <b>Yes</b>"]
        E --> D2["Embed in closed product? <b>Yes</b>"]
        E --> D3["Must changes stay open? <b>No</b> (optional)"]
        F --> W1["Run & Modify? <b>Yes</b>"]
        F --> W2["Embed in closed product? <b>Yes, with conditions</b>"]
        F --> W3["Must changes stay open? <b>Only the covered file or library</b>"]
        D1 --> E1["Run & Modify? <b>Yes</b>"]
        D2 --> E2["Embed in closed product? <b>No</b>"]
        D3 --> E3["Must changes stay open? <b>Yes</b> (mandatory)"]
        E1 --> G
        E2 --> G
        E3 --> G
    end
    G["<b>Reciprocity only triggers on distribution</b><br>Running modified code internally creates no obligation"]
    style box fill:#e6ffe6,stroke:#2b8a3e,stroke-width:2px,color:#1b4332;
    classDef green fill:#e6ffe6,stroke:#2b8a3e,stroke-width:2px,color:#1b4332;
    classDef red fill:#ffe3e3,stroke:#e03131,stroke-width:2px,color:#5c0000;
    classDef yellow fill:#ffff9d,stroke:#f59f00,stroke-width:2px,color:#5c3c00;
    classDef amber fill:#fff3bf,stroke:#f08c00,stroke-width:2px,color:#5c3c00;
    classDef white fill:#f8f9fa,stroke:#adb5bd,stroke-width:2px,color:#212529;
    classDef dashed fill:#f8f9fa,stroke:#adb5bd,stroke-width:2px,stroke-dasharray: 5 5,color:#000000;
    classDef dashed_red fill:#ffe3e3,stroke:#adb5bd,stroke-width:2px,stroke-dasharray: 5 5,color:#000000;
    classDef note fill:#ffffff,stroke:#868e96,stroke-width:1px,stroke-dasharray: 3 3,color:#212529;
    classDef box_fill fill:#ffffff,stroke:#adb5bd,stroke-width:1px;
    class D1,D2,D3,W1,E1 green;
    class W2,W3 amber;
    class E2 red;
    class E3 green;
    class D yellow;
    class W amber;
    class E yellow;
    class F dashed;
    class B dashed_red;
    class A,C white;
    class G note;
    class box box_fill;
```

Copyright Foundation: Expression vs. Ideas

Under Directive 2009/24/EC, software is protected by copyright as a **literary work** (Art. 1(1)). But copyright protects only the **expression**, not the ideas beneath it: Art. 1(2) explicitly excludes “ideas and principles which underlie any element of a computer program, including those which underlie its interfaces.”

- **Protected:** your specific source code text, binaries, container recipes, prompt text, and preparatory design material.
- **Not protected:** mathematical algorithms, scientific models, programming logic, data structures, and interfaces.

The CJEU confirmed this line in *SAS Institute v World Programming* (C-406/10): a program’s functionality, its programming language, and its data file formats are ideas, not expression, and are therefore outside copyright. Someone may reimplement your algorithm from scratch; they may not copy your code. This is exactly why licenses exist — they set the terms for the expression, which is the only part copyright lets you control.

Scope of this Lesson: What Counts as *Software*?

Across international frameworks (17 U.S.C. § 101 and WIPO model provisions), software is broadly defined as a set of statements or instructions used directly or indirectly in a computer to bring about a certain result. Research software goes well beyond Python scripts, so this lesson covers six asset types — find the ones matching your own project, since the scenarios later map onto them:

- **Source Code** — original algorithms, or implementations of published methods.
- **Third-Party Integrations** — embedded snippets and linked libraries (static or dynamic).
- **Infrastructure as Code** — Ansible playbooks, Terraform configs, container recipes (`Dockerfile`, Apptainer `.def`).
- **Container Images** — built binary snapshots (`.sif` files, OCI registry images).
- **AI-Assisted Code** — generated or refactored with human oversight.
- **AI Prompt Templates** — engineered system prompts meeting the threshold of human authorship.

Motivation: Debugging a License Compliance Failure

With the three license families in mind, examine what happens when they collide inside an automated CI/CD pipeline:

```
%%{init: {themeVariables: { 'edgeLabelBackground': '#faf5ff' }}}%% flowchart TB
subgraph CI_CD_License_Compliance_Debugging_Pipeline [CI/CD License Compliance Debugging Pipeline]
    A["Paste a snippet copied from somewhere"] --> B["Run Compliance Scanner"]
    B --> C["Check Inbound vs. Outbound Terms"]
    C --> D["Target license: Permissive<br/>Pasted snippet: Copyleft"]
    D --> E["Select Patch Option"]
    E --> F["Keep MIT, add comment '# Originally GPL'"]
    F --> G["Re-license repo to GPL-3.0 / EUPL-1.2"]
    G --> H["Reimplement the functionality yourself"]
    H --> I["Delete LICENSE file to silence the scanner"]
end
```

Diagram illustrating the CI/CD License Compliance Debugging Pipeline:

- A: Paste a snippet copied from somewhere
- B: Run Compliance Scanner
- C: Check Inbound vs. Outbound Terms
- D: Target license: Permissive
Pasted snippet: Copyleft
- E: Select Patch Option
- F: Keep MIT, add comment '# Originally GPL'
- G: Re-license repo to GPL-3.0 / EUPL-1.2
- H: Reimplement the functionality yourself
- I: Delete LICENSE file to silence the scanner

Flowchart labels and status indicators:

- D: **✗** BUILD FAILURE
- F: **✗** BUILD FAIL
- G: **✓** BUILD PASS
- H: **✓** BUILD PASS
- I: **⚠** SCANNER PASSES

LEGAL TRAP
Still infringing, and your own code reverts to All Rights Reserved"] P["Permissive
MIT, Apache-2.0, 0BSD"] WC["Weak Copyleft
LGPL, MPL-2.0, EPL-2.0"] CL["Copyleft
GPL-3.0, EUPL-1.2"] end P -.->|"What I want for my repo"| C CL -.->|"What the pasted snippet uses"| C WC -.->|"Would often have been fine"| C classDef pass fill:#e6ffe6,stroke:#2b8a3e,stroke-width:2px,color:#1b4332; classDef copyleft fill:#fff9db,stroke:#f59f00,stroke-width:2px,color:#5c3c00; classDef amber fill:#fff3bf,stroke:#f08c00,stroke-width:2px,color:#5c3c00; classDef fail fill:#ffe3e3,stroke:#e03131,stroke-width:2px,color:#5c0000; classDef warning fill:#fff3bf,stroke:#f08c00,stroke-width:2px,color:#5c0000; classDef neutral fill:#f8f9fa,stroke:#495057,stroke-width:2px,color:#212529; classDef box_fill fill:#ffffff,stroke:#adb5bd,stroke-width:1px; class P,G,H pass; class CL copyleft; class WC amber; class D,F fail; class I warning; class A,A2,B,C,E neutral; class box box_fill;

- Option D is the one worth dwelling on: deleting the **LICENSE** file makes the scanner quiet without changing anything legally. You are still distributing someone else's copyleft code without honouring its terms, and you have now stripped your own users of any permission to use your work. A green pipeline is not a compliance result.
- Option C works only if you genuinely reimplement the functionality without copying the original expression. As the idea/expression split above establishes, the algorithm is free to reuse — the specific code is not. Reading the original closely and retyping a close paraphrase is still copying.

Limitations of AI-Assisted Licensing Advice

Modern software developers and RSEs routinely rely on AI coding assistants (ChatGPT, Claude, GitHub Copilot) to generate boilerplate, refactor functions, and answer project setup questions.

However, using these tools for legal or licensing guidance introduces a subtle risk of **AI legal bias**. AI models are overwhelmingly trained on US-centric web data and legal forum posts, so their outputs default almost universally to **US common law concepts** such as *Fair Use*, *Work Made for Hire*, and *Derivative Works*.

Developers working under EU statutory frameworks face a different legal reality around exceptions, ownership, and code adaptation. The clearest example is the term you will hear constantly:

- **US law (17 U.S.C. § 101)** formally defines “*derivative work*”, and AI assistants reach for it to describe almost any code modification.
- **EU law (Directive 2009/24/EC, Art. 4(1)(b))** does not use that term at all. It grants exclusive rights over “the translation, adaptation, arrangement and any other alteration of a computer program” — governed collectively as an **adaptation**.
- **Licenses use it anyway:** **GPL-3.0** and **EUPL-1.2** define “derivative work” inside their own text as a contractual term for international enforceability, even though EU statute treats the act as an adaptation.

So when an AI assistant tells you a snippet creates a “derivative work”, treat that as a prompt to check the actual question under EU law: is this a statutory **adaptation**, or a **combined work** across a technical boundary? The rest of this lesson gives you that EU-aligned

framework.

Standardizing In-File Declarations: SPDX Identifiers

Selecting a license is only half the job. Automated scanners and CI/CD pipelines need a machine-readable way to verify compliance per file without parsing legal text.

Managed by the Linux Foundation, **SPDX identifiers** are standardized short tags (`MIT` , `Apache-2.0` , `GPL-3.0-only` , `EURL-1.2`) placed at the top of every source file:

```
# SPDX-License-Identifier: MIT
# Copyright (c) 2026 Author Name <author@institute.eu>
```

Every scenario below shows the SPDX tagging for its asset type — Python scripts, container recipes, and prompt templates each have their own conventions.

License Selection Decision Matrix & Scenario Index

Our decision framework is grounded in the European Commission’s [Joinup Licensing Assistant \(JLA\)](#) , which sorts licenses across six criteria: **Can** (permissions), **Must** (obligations), **Cannot** (restrictions), **Compatible** (interoperability), **Law** (jurisdiction), and **Support** (governance).

The scenarios below are independent. Find the row that matches what you are actually building, jump to it, and skip the rest.

If you are...	Scenario	Typical Outcome	Example Licenses
Writing everything yourself	1. Own code	● Free choice	<code>MIT</code> , <code>Apache-2.0</code> , <code>BSD-3-Clause</code>
Implementing a published algorithm	2. Algorithm implementation	● Free choice — copyleft if you want reciprocity	<code>EURL-1.2</code> , <code>GPL-3.0</code> , <code>AGPL-3.0</code>
Pasting in a permissive snippet	3. Embed permissive	● Stay permissive, keep notices	<code>MIT</code> , <code>Apache-2.0</code> , <code>EURL-1.2</code> , <code>GPL-3.0</code>
Pasting in a copyleft snippet	4. Embed copyleft	● Strong copyleft likely required	<code>EURL-1.2</code> , <code>GPL-3.0</code>
Importing or linking a library	5. Link a library	● Depends on which copyleft — see below	<code>GPL-3.0</code> , <code>EURL-1.2</code> , or permissive if weak copyleft
Writing a Dockerfile or <code>.def</code>	6. Container recipe	● Free choice	<code>MIT</code> , <code>Apache-2.0</code> , <code>BSD-3-Clause</code>
Publishing a built image	7. Built image	⚠ Multi-license bundle	Governed by each layer’s own terms

If you are...	Scenario	Typical Outcome	Example Licenses
Using Copilot, ChatGPT or Claude	8. AI-assisted code	● Free choice, verify for memorization	<code>MIT</code> , <code>Apache-2.0</code> , <code>EUPL-1.2</code> , <code>GPL-3.0</code>
Shipping prompts, weights or datasets	9. AI workflows & assets	● Dual-license code vs. assets	<code>MIT</code> + <code>CC-BY-4.0</code>

This lesson covers nine scenarios, a typical session works through three or four. The rest are here for reference when your project changes

Module 1: Clean Slate – Authoring Original Code & Algorithms

When writing original code or implementing published algorithms, no third-party license constrains your choice — but who owns the code depends on your employment contract and national rules, so check your institution’s policy first.

Scenario 1: Authoring original code and algorithms

You wrote an original algorithm from scratch (in Python, C++, Rust, etc.). Your repository contains only your original source code and dependency specifications

(`requirements.txt`, `CMakeLists.txt`, `Cargo.toml`).

- **Licensing Goal:** You want **maximum adoption** and zero friction for commercial or academic reuse.
- **Legal Reality:** External dependencies remain separate works. Because you have not bundled third-party code inside your repository, no inbound license terms constrain your choice.
- **JLA Selection Strategy:** To ensure downstream users must acknowledge your original authorship while granting them maximum flexibility to incorporate your code into both open and proprietary software, you require citation credit (`Incl. Copyright`) without imposing share-alike conditions (leaving `Copyleft/Share a.` unselected).

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select `Distribute`, `Modify/merge`, and `Commercial use`
2. **Must Column:** Select `Incl. Copyright`
3. **Support Column:** Select `OSI approved`

- **Example JLA Matches:** `MIT`, `Apache-2.0`, `BSD-3-Clause`
- **Permissive vs. Public Domain (EU Civil Law Nuance):** Public domain dedications (e.g., `CC0`, `Unlicense`) attempt to give away all rights. However, under EU civil law, authors cannot legally give up their moral rights (*droit moral*). Selecting an

explicit permissive license like `MIT` or `Apache-2.0` grants broad permissions globally, remains legally valid under European copyright law, and guarantees academic citation credit.

- **Downstream Obligations:** Anyone who reuses, modifies, or integrates your code into their work must preserve your copyright notice and license text. They are not required to share their modifications or open-source their downstream projects.
- **Allowed Inbound Snippets:** If you want to include small third-party code snippets in your files, you can freely embed code licensed under **permissive terms** (e.g., MIT, BSD, Apache-2.0, 0BSD) or public domain waivers (CC0) without affecting your permissive license. However, embedding copyleft snippets (e.g., GPL, EUPL) might trigger reciprocal obligations requiring you to re-license. Whether it does depends on which copyleft: weak copyleft (LGPL, MPL-2.0) often lets your surrounding code stay permissive, while strong copyleft generally does not.
- **In-File Identification (SPDX):** Apply standard machine-readable SPDX identifier comments directly at the top of your scripts:

```
# SPDX-License-Identifier: MIT
# Copyright (c) 2026 Author Name <author@institute.eu>

import numpy as np
```

Scenario 2: Choosing reciprocity for your own implementation

You developed a custom solver implementing algorithms from academic literature. You want any downstream improvements, extensions, or modifications to remain open-source and be shared back with the scientific community.

- **Licensing Goal:** You want to enforce **reciprocity** (share-alike), preventing third parties from incorporating your implementation into proprietary software without sharing their modifications.
- **Legal Reality:** The published algorithm itself is an unprotected idea — anyone may implement it independently, as Scenario 1 and the SAS ruling establish. What copyright protects is *your* specific implementation. Nothing about implementing a published method forces a particular license; copyleft here is your deliberate choice to bind downstream distributors to matching terms.
- **JLA Selection Strategy:** To enforce reciprocal sharing, you must mandate that downstream distributors disclose their modified source code (`Disclose source`) and license their adaptations or combined works under matching terms (`Copyleft/Share a.`).

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select `Distribute` , `Modify/merge` , and `Commercial use`
2. **Must Column:** Select `Incl. Copyright` , `Disclose source` , and `Copyleft/Share a.`
3. **Support Column:** Select `OSI approved`

- **Example JLA Matches:** `EUPL-1.2` , `GPL-3.0` , `AGPL-3.0`
- **Copyleft Mechanics (EUPL vs. GPL Nuance):** `GPL-3.0` is the standard global copyleft license, but `EUPL-1.2` is specifically tailored for European institutions. EUPL-1.2 is officially published in 23 EU language versions (each with equal legal validity), includes built-in compatibility clauses with GPL, and explicitly defaults to EU Member State jurisdiction and courts.
- **A caution before choosing strong copyleft:** reciprocity also limits who can combine with your code. Strong copyleft licenses are frequently incompatible with each other, so a future collaborator on a differently-licensed copyleft project may be unable to use your work at all. Scenario 5 covers this.
- **Downstream Obligations:** Anyone who distributes your code or a modified version of it must provide complete access to the corresponding source code under the same copyleft license and preserve your original copyright notices.
- **Allowed Inbound Snippets:** You can freely embed code snippets licensed under **permissive terms** (e.g., MIT, Apache-2.0, BSD) or public domain waivers (CC0). You may also embed snippets from compatible copyleft code (e.g., EUPL, GPL). However, you cannot embed closed-source or proprietary code snippets.
- **In-File Identification (SPDX):** Apply standard machine-readable SPDX identifier comments directly at the top of your scripts:

```
# SPDX-License-Identifier: EUPL-1.2
# Copyright (c) 2026 Author Name <author@institute.eu>

import numpy as np
```

Module 2: The Dependency Minefield – Inbound Code & Linking

Embedding third-party snippets or linking against external libraries introduces boundaries that can constrain your license choice. How far those boundaries reach depends on which license the inbound code carries.

Scenario 3: Embedding permissively licensed third-party code

You are building an RSE tool and copied a helper function or utility snippet from a third-party project licensed under a permissive license (e.g., MIT or Apache-2.0) directly into one of your source files.

- **Licensing Goal:** You want to maintain a **permissive default** for your project while properly acknowledging and legally respecting the embedded third-party code.

- **Legal Reality:** Permissive licenses explicitly grant you permission to copy, modify, and embed their code into your repository. However, embedding permissive code does not make the original third-party copyright disappear, you must preserve the original copyright attribution and license terms for that specific snippet.
- **JLA Selection Strategy:** Because inbound permissive code gives you maximum licensing flexibility, your overall repository can remain permissively licensed. To reflect this, select citation obligations (`Incl. Copyright`) without imposing reciprocal sharing constraints (leaving `Copyleft/Share a.` unselected).

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select `Distribute` , `Modify/merge` , and `Commercial use`
 2. **Must Column:** Select `Incl. Copyright`
 3. **Support Column:** Select `OSI approved`
- **Example JLA Matches:** `MIT` , `Apache-2.0` , `BSD-3-Clause`
 - **Notice Preservation Nuance:** Permissive licenses are flexible, but they are not license-free. If you copy code from an Apache-2.0 or BSD-3-Clause project into your MIT-licensed repository, you must retain the original author's copyright statement and license identifier directly above the embedded code block. Your repository license covers *your* code. It does not relicense the embedded snippet — that code stays under its original license and its original copyright holder's terms. You are distributing one file containing two separately licensed contributions, which is why both notices must appear.
 - **Downstream Obligations:** Downstream users receive your project under your primary permissive license (e.g., MIT), but they must preserve both your overall copyright notice and the specific third-party notices attached to embedded snippets.
 - **Allowed Inbound Snippets:** In addition to the embedded permissive snippet, you can freely embed other permissively licensed code (MIT, BSD, Apache-2.0) or public domain waivers (CC0). Embedding **strong** copyleft code (GPL, EUPL) generally requires re-licensing your repository to match. Weak copyleft (LGPL, MPL-2.0) applies at a narrower boundary and often does not
 - **In-File Identification (SPDX):** Mark both your overall file license and the specific embedded snippet using SPDX comments:

```
# SPDX-License-Identifier: MIT
# Copyright (c) 2026 Author Name <author@institute.eu>

# --- Embedded Third-Party Snippet ---
# SPDX-License-Identifier: Apache-2.0
# Copyright (c) 2024 External Contributor <dev@external-lib.org>
def fast_matrix_solver(matrix):
    # Embedded algorithm implementation
    return np.linalg.solve(matrix, np.eye(len(matrix)))
# --- End Embedded Snippet ---

def main():
    pass
```

Scenario 4: Embedding copyleft third-party code

You are building a software tool and copied a non-trivial code snippet from a third-party project licensed under a copyleft license (e.g., GPL-3.0 or EUPL-1.2) directly into one of your source files.

- **Licensing Goal:** Comply with legal requirements imposed by the inbound copyleft code while ensuring your overall repository remains legally compliant.
- **Legal Reality:** Copying a non-trivial copyleft snippet into your source files creates a single combined work, so copyleft licensing generally extends to your whole project. “Non-trivial” matters: a snippet too short or purely functional to qualify as the author’s own intellectual creation (Art. 1(3)) may not carry copyright at all. There is no word count or line count that draws this line — if you are unsure, assume it is protected and either comply or reimplement.
- **JLA Selection Strategy:** Because the inbound copyleft code forces your repository to adopt reciprocal sharing terms, you must configure JLA to require source code disclosure (**Disclose source**) and reciprocal licensing (**Copyleft/Share a.**).

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select **Distribute** , **Modify/merge** , and **Commercial use**
2. **Must Column:** Select **Incl. Copyright** , **Disclose source** , and **Copyleft/Share a.**
3. **Support Column:** Select **OSI approved**

- **Example JLA Matches:** **GPL-3.0** , **EUPL-1.2**
- **Copyleft Scope & EUPL Compatibility (Legal Nuance):** Directly copying copyleft code into your source files extends the copyleft obligation to your entire codebase. If the embedded snippet is **EUPL-1.2** , its built-in compatibility provisions allow you to license your combined project under **GPL-3.0** if your project ecosystem requires it, resolving license conflicts without violating EUPL terms.

- **Downstream Obligations:** Anyone who receives, modifies, or distributes your repository must receive full access to the source code under the same copyleft license terms ([GPL-3.0](#) or [EURL-1.2](#)) and preserve all copyright notices.
- **Allowed Inbound Snippets:** Because your overall repository is now governed by a copyleft license, you can safely embed code from **permissive sources** (MIT, BSD, Apache-2.0, CC0) as well as **compatible copyleft sources**. You cannot embed proprietary, closed-source code or snippets from incompatible copyleft licenses.
- **In-File Identification (SPDX):** Mark your overall file license and clearly cite the embedded copyleft snippet using SPDX comments:

```
# SPDX-License-Identifier: GPL-3.0-or-later
# Copyright (c) 2026 Author Name <author@institute.eu>

# --- Embedded Copyleft Snippet ---
# SPDX-License-Identifier: GPL-3.0-or-later
# Copyright (c) 2023 External Researcher <researcher@university.org>
def optimized_fft_filter(data_signal):
    # Embedded copyleft algorithm implementation
    return np.fft.fft(data_signal)
# --- End Embedded Snippet ---

def main():
    pass
```

Module 3: Dependency Linking & Packaging

When software incorporates external dependencies, whether by dynamic linking, static compiling, or bundling binaries into container images licensing obligations expand beyond your own written source code. This module covers how dependency boundaries, build automation scripts, and packaged container artifacts affect legal compliance under the Joinup Licensing Assistant (JLA) framework.

Scenario 5: Linking against a GPL-licensed library

You are developing a software application that imports or links against an external software library licensed under GPL-3.0 (e.g., importing a GPL Python package or linking a C/C++ static/shared library).

- **Licensing Goal:** Ensure legal compliance while using copyleft libraries as core dependencies in your software project.
- **Legal Reality:** Whether linking creates a combined work is genuinely unsettled, and often has to be decided case by case. The FSF's position is that linking a GPL library — statically or dynamically — creates a combined work; some legal scholars and Commission EURL guidance disagree, particularly for dynamic linking through a stable API. Most Member States have no case law on this, so no firm general rule can be stated. The guidance below follows the conservative, widely-adopted reading.

- **JLA Selection Strategy:** Because linking to a GPL library requires your distributed project to be released under matching reciprocal terms, you must configure JLA to mandate source code disclosure (`Disclose source`) and reciprocal licensing (`Copyleft/Share a.`).

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select `Distribute` , `Modify/merge` , and `Commercial use`
2. **Must Column:** Select `Incl. Copyright` , `Disclose source` , and `Copyleft/Share a.`
3. **Support Column:** Select `OSI approved`

- **Example JLA Matches:** `GPL-3.0` , `EURL-1.2`
- **Linking Boundaries & License Selection (Legal Nuance):**
 - **Why GPL forces copyleft:** Linking against a standard `GPL-3.0` library extends copyleft to your entire project. Your repository must adopt a compatible copyleft license (`GPL-3.0` or `EURL-1.2` , which explicitly lists GPL-3.0 in its compatibility appendix).
- **Copyleft licenses are not compatible with each other:** two strong copyleft licenses can each demand that the combined work use *their* terms, which makes the combination undistributable. The classic trap is `GPL-2.0-only` : without the “or later” clause you cannot upgrade to GPL-3.0 to resolve a conflict, so GPL-2.0-only code cannot be combined with GPL-3.0 or Apache-2.0 code at all. Always check the exact SPDX identifier — `GPL-2.0-only` and `GPL-2.0-or-later` behave very differently.
- **Downstream Obligations:** Anyone to whom you **distribute** the application must receive full access to your source code under `GPL-3.0` (or `EURL-1.2`), along with upstream copyright notices and the build scripts needed to recompile it. Running the software internally, without distributing it, creates no such obligation — though note that `AGPL-3.0` extends this to network use.
- **Allowed Inbound Code & Dependencies:** Your project can import or include other **permissively licensed** packages (MIT, BSD, Apache-2.0) and public domain waivers (CC0). However, all code linked together in the final executable or runtime environment must satisfy GPL compatibility.
- **In-File Identification (SPDX):** Apply standard machine-readable SPDX identifier comments directly at the top of your main scripts:

```
# SPDX-License-Identifier: GPL-3.0-or-later
# Copyright (c) 2026 Author Name <author@institute.eu>

import gpl_licensed_solver # External GPL dependency forces GPL/EURL compliance

def solve_system(data):
    return gpl_licensed_solver.compute(data)
```



Scenario 6: Authoring container recipes and environment specifications

You are creating a `Dockerfile`, Conda `environment.yml`, or build recipe to automate the setup of your research environment. The recipe itself contains setup instructions, shell commands, and package lists.

- **Licensing Goal:** You want **maximum adoption** and reuse of your build automation script so other researchers can freely adapt and build upon your workflow.
- **Legal Reality:** Build recipes and configuration scripts are plain-text source code, separate from the software binaries they download at build time. The build instructions you write are your expression — but note that a very short recipe (a `FROM` line plus two `RUN` commands) may be too trivial to meet the Art. 1(3) originality threshold and may not attract copyright at all. Longer, non-obvious recipes clearly do.
- **JLA Selection Strategy:** To allow anyone to reuse or adapt your container recipe without restrictions, you require citation credit (`Incl. Copyright`) while leaving reciprocal requirements (`Copyleft/Share a.`) unselected.

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select `Distribute`, `Modify/merge`, and `Commercial use`
2. **Must Column:** Select `Incl. Copyright`
3. **Support Column:** Select `OSI approved`

- **Example JLA Matches:** `MIT`, `Apache-2.0`, `BSD-3-Clause`
- **Recipe vs. Image Nuance:** The license applied to a `Dockerfile` covers only the recipe instructions, not the software packages installed inside the container when `docker build` runs. A permissively licensed Dockerfile can install both permissive and copyleft packages without legal conflict.
- **Downstream Obligations:** Anyone who reuses or adapts your build recipe must preserve your original copyright notice in the header of the recipe file.
- **Allowed Inbound Snippets:** You can freely include build commands and code snippets from permissively licensed build scripts or public domain code.
- **In-File Identification (SPDX):** Place SPDX identifier comments at the top of your Dockerfile or recipe file:

```
# SPDX-License-Identifier: MIT
# Copyright (c) 2026 Author Name <author@institute.eu>

FROM ubuntu:24.04
RUN apt-get update && apt-get install -y python3 python3-pip
COPY solver.py /app/solver.py
```

Scenario 7: Distributing pre-built container images

You compiled and published a pre-built container image (e.g., pushing a compiled Docker image to Docker Hub, GitHub Container Registry, or an institutional registry) containing an OS layer, runtime binaries, dependencies, and your application code.

- **Licensing Goal:** Safely distribute compiled container images without violating the license terms of any software layer or binary included inside the image.
- **Legal Reality:** A compiled container image is a **multi-license aggregate bundle**, not a single combined work. Distributing pre-built binaries makes you a distributor of every package inside, so source-availability obligations apply to the copyleft components (Linux base packages, coreutils, GPL libraries). But those packages sitting in the same filesystem as your application do not make your application a derivative of them – this is mere aggregation. Your own code keeps whatever license you chose; you simply also carry distributor obligations for the copyleft software you are shipping alongside it.
- **JLA Selection Strategy:** Because a container image combines multiple distinct software components, JLA is used to evaluate constituent component obligations. When distributing compiled binaries containing copyleft layers, source disclosure requirements (`Disclose source`) must be fulfilled for those specific layers.

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select `Distribute` and `Commercial use`
2. **Must Column:** Select `Incl. Copyright` and `Disclose source`
3. **Support Column:** Select `OSI approved`

- **JLA Outcome:** No single license applies. Use JLA per component to check each one's obligations, then record the aggregate in your image metadata.
- **Multi-License Aggregation Nuance:** Applying a permissive license (like MIT) to your application code inside the container does not override or erase the GPL/LGPL obligations of base system packages installed in `/usr/lib` or `/usr/bin` . Distributing the built image binary makes you a distributor of all installed packages.

- **Downstream Obligations:** You must ensure downstream users can obtain the corresponding source for the copyleft components you shipped. Publishing your `Dockerfile` documents the build but does not by itself satisfy this — the GPL asks for the source of the binaries actually distributed. In practice, most research images rely on unmodified upstream distribution packages, where pointing to the distributor's public source archives (as GPLv3 §6(d) permits) is the normal approach. If you modify or rebuild a copyleft component yourself, you must provide that source directly.
- **Allowed Inbound Packages:** Before publishing an image binary, run automated compliance scanning tools (e.g., Syft, Trivy) to generate a Software Bill of Materials (SBOM) and verify that no non-redistributable or proprietary software is packaged inside.
- **In-File Identification (Metadata Annotations):** Document the multi-license nature of the aggregate bundle using standard OCI (Open Container Initiative) image labels inside your Dockerfile:

```
# SPDX-License-Identifier: MIT
# Copyright (c) 2026 Author Name <author@institute.eu>

FROM ubuntu:24.04
LABEL org.opencontainers.image.authors="author@institute.eu"
# OCI Standard Image Annotations for Docker Hub Compliance
LABEL org.opencontainers.image.title="My Research Pipeline"
LABEL org.opencontainers.image.licenses="MIT AND GPL-3.0-or-later"
LABEL org.opencontainers.image.vendor="My Institute Name"
LABEL org.opencontainers.image.description="Includes Ubuntu 24.04 base layers (GPL/LGPL) and custom solver (MIT)"

COPY solver.py /app/solver.py
```

Module 4: Emerging Workflows & AI

AI-assisted development tools and machine learning models introduce unique legal challenges regarding copyright ownership, training data memorization, and behavioral restrictions. This module addresses how to license projects built with AI code generation tools and how to package research software that bundles AI models, weights, and datasets alongside source code.

Scenario 8: AI-assisted code generation

You used AI tools (e.g., GitHub Copilot, ChatGPT, Claude) to write functions, unit tests, or documentation for your research software repository.

- **Licensing Goal:** Retain clear ownership and apply a **permissive license** (`MIT` or `Apache-2.0`) to your repository without incurring hidden copyright infringement or copyleft obligations from code embedded during model training.

- **Legal Reality:** Unmodified AI-generated outputs lack human authorship and are generally not eligible for copyright protection under current EU and international legal standards. However, if an LLM reproduces a substantial copyrighted code snippet verbatim from its training data (memorization), that output snippet retains its original copyright and license obligations.
- **JLA Selection Strategy:** To ensure maximum adoption and academic reuse for your overall codebase, require citation credit (`Incl. Copyright`) while avoiding share-alike constraints (leaving `Copyleft/Share a.` unselected), supported by automated compliance checks.

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select `Distribute` , `Modify/merge` , and `Commercial use`
2. **Must Column:** Select `Incl. Copyright`
3. **Support Column:** Select `OSI approved`

- **Example JLA Matches:** `MIT` , `Apache-2.0` , `BSD-3-Clause`
- **Marking AI-generated code:** Some projects and AI tool terms require contributors to disclose AI involvement — via a commit trailer, a PR checkbox, or an in-file comment. Even where it is optional, marking AI-assisted sections is increasingly recommended practice: it records provenance, signals to reviewers where extra scrutiny is warranted, and makes later authorship or infringement questions much easier to resolve. Check the contribution guidelines of any project you submit to.
- **Downstream Obligations:** Downstream users must preserve your copyright notice for the repository. They are free to reuse, modify, and integrate your code into commercial or open-source projects.
- **Allowed Inbound Snippets:** You can include permissively licensed code, public domain code (CC0), and AI-generated snippets that have been verified against verbatim training data duplication.
- **In-File Identification (SPDX):** Apply standard machine-readable SPDX identifier comments directly at the top of your scripts:

```
# SPDX-License-Identifier: MIT
# Copyright (c) 2026 Author Name <author@institute.eu>

def filter_sensor_data(raw_readings: list[float]) -> list[float]:
    """Cleans raw sensor data (written with AI assistance and human review)."""
    return [reading for reading in raw_readings if reading > 0.0]
```

You are developing research software that includes source code alongside trained machine learning model weights (`.pt` , `.safetensors`) and benchmark datasets.

- **Licensing Goal:** Apply a clear **dual-licensing strategy** that makes both the software source code and the non-code assets (data, weights) open and reusable under appropriate legal frameworks.
- **Legal Reality:** Standard software licenses (MIT, GPL) are written for source code and fit datasets and model parameters poorly. Datasets may attract the EU *sui generis* database right where there has been substantial investment in obtaining, verifying, or presenting their contents. Model weights are a harder case: they are neither code nor a database, and whether they attract any copyright protection in the EU is genuinely unsettled. Because of this uncertainty, applying an explicit license to weights is about setting clear terms for your users, not about relying on a settled legal right.
- **JLA Selection Strategy:** Use JLA to select an OSI-approved open-source license for the executable code component (`Incl. Copyright` selected), while using Creative Commons licenses (e.g., `CC-BY-4.0` or `CC0`) for the dataset and weight files.

✓ Solution

What to select in the JLA interface:

1. **Can Column:** Select `Distribute` , `Modify/merge` , and `Commercial use`
2. **Must Column:** Select `Incl. Copyright`
3. **Support Column:** Select `OSI approved`

- **Example JLA Matches:** `MIT` , `Apache-2.0` (for the code component)
- **Code vs. Data/Weights & OpenRAIL Nuance:** Avoid applying software licenses like GPL or MIT to raw datasets or model weights — their terms reference source code, object code, and linking, which leaves users guessing about what applies. Use **CC-BY-4.0** or **CC0** for non-code assets instead. Note also that behavioral licenses (such as OpenRAIL) impose usage restrictions (e.g., prohibiting specific harmful uses), so they do **not** qualify as OSI-approved open source and will not appear in standard JLA queries.
- **Downstream Obligations:** Downstream users must cite your repository for the code (under your chosen software license) and give credit for the model weights and data under the corresponding Creative Commons license.
- **Allowed Inbound Assets:** You may combine permissively licensed python code with CC-BY-4.0 datasets or open-weight models, provided the attribution files clearly separate code licenses from data/weight licenses.
- **In-File Identification (SPDX / Dual-Licensing Structure):** Document the dual-licensing scheme in your root repository structure and script headers:

```
# SPDX-License-Identifier: MIT
# Copyright (c) 2026 Author Name <author@institute.eu>
#
# Note: Source code is licensed under MIT.
# Model weights in /models/ and datasets in /data/ are licensed under CC-BY-4.0.

import torch

def load_pipeline():
    model = torch.load("models/climate_weights.safetensors")
    return model
```

Best Practices: Attaching a License to Your Repository

Once you have selected a license using the JLA, you must officially attach it to your repository so automated scanners, package registries, and downstream users can verify your terms.

1. Adding the Root `LICENSE` File

Always place the full text of your chosen license in a plain-text file named `LICENSE` or `LICENSE.txt` at the root of your repository.

- **Exact Legal Text:** Copy the standard text directly from spdx.org/licenses or choosealicense.com.
- **Copyright Header:** Ensure you fill in the copyright year and copyright holder line at the top of the license text:

```
Copyright (c) 2026 [Author Name or Institution Name]
```

- **Do Not Edit Terms:** Never modify the legal wording of standard licenses (e.g., removing clauses from GPL or MIT). Edited texts are no longer the license they claim to be: compliance scanners cannot classify them, package registries may flag them, and downstream users have to get their own legal review before touching your code. If a standard license does not fit, pick a different standard license.

2. Documenting License Status in `README.md`

Add a dedicated **License** section near the bottom of your repository's `README.md` file, along with a machine-readable badge:

License

This project is licensed under the MIT License - see the [**LICENSE**](LICENSE) file for details.

[![License: MIT](https://img.shields.io/badge/License-MIT-yellow.svg)](https://opensource.org/licenses/MIT)

3. Automated Compliance with the REUSE Standard

For multi-asset research repositories containing code, data, container build recipes, and prompt templates, follow the [FSFE REUSE Initiative](#) standard:

1. **Include License Texts:** Place full license files inside a `LICENSES/` directory (e.g., `LICENSES/MIT.txt`, `LICENSES/GPL-3.0-or-later.txt`).
2. **Add In-File SPDX Headers:** Label every source file, build script, and prompt file with SPDX tags.
3. **Verify Compliance:** Run the automated REUSE linter in your CI/CD pipeline:

```
# Install and run REUSE compliance check
pip install reuse
reuse lint
```

When `reuse lint` passes, downstream researchers can automatically verify the legal status of every single asset in your codebase.

Summary: Resolving the Compliance Pipeline

When developing research software, license compliance is not an afterthought to debug at the end of a project, it is a proactive design choice. By using the **Joinup Licensing Assistant (JLA)** framework to align your repository license with your inbound dependencies from day one, your CI/CD pipeline passes cleanly on the first run.

The diagram below illustrates how selecting a compatible license upfront ensures your code passes automated compliance checks and results in a legally sound release:

```
%%{init: {'themeVariables': { 'edgeLabelBackground': '#faf5ff' }}}%% flowchart TB
    subgraph local ["1. Local Authoring & Standardization"]
        A["<b>Inbound Reuse Trigger:</b><br>User copies copyleft snippet <i>(Scenario 4)</i><br>or links GPL library <i>(Scenario 5)</i>"] --> B["<b>JLA Selection Strategy:</b><br>Select compatible copyleft license<br><i>(GPL-3.0 / EUPL-1.2)</i>"]
        B --> C["<b>Standardize Local Codebase:</b><br>1. Add <b>SPDX Headers</b> to all files <i>(Scenarios 1-9)</i><br>2. Add root <code>LICENSE</code> file & README badge"]
    end
    subgraph cicd ["2. Automated CI/CD & Verification"]
        C --> D["<b>Git Push</b> to Repository"]
        D --> E["<b>Build Trigger: Run Compliance Scanner</b><br><i>(Executes <code>reuse lint</code> in CI/CD)</i>"]
        E --> F["<b>Verify Inbound vs. Outbound Terms</b>"]
        F --> G["<b>SPDX Headers & License Match Confirmed!</b>"]
        G --> H["<b>BUILD PASSES</b><br>Compliance verified automatically"]
        H --> I["SUCCESS"]
    end
    I --> J["<b>COMPLIANT OPEN-SOURCE RELEASE</b><br>Legally sound, reproducible & ready for"]
```

```
scientific reuse"] end classDef pass fill:#e6ffe6,stroke:#2b8a3e,stroke-  
width:2px,color:#1b4332; classDef neutral fill:#f8f9fa,stroke:#495057,stroke-  
width:2px,color:#212529; classDef box_fill fill:#ffffff,stroke:#adb5bd,stroke-width:1px; class  
F,SUCCESS pass; class A,B,C,D,E neutral; class local,cicd box_fill;
```

Scenario Mapping Across the Pipeline

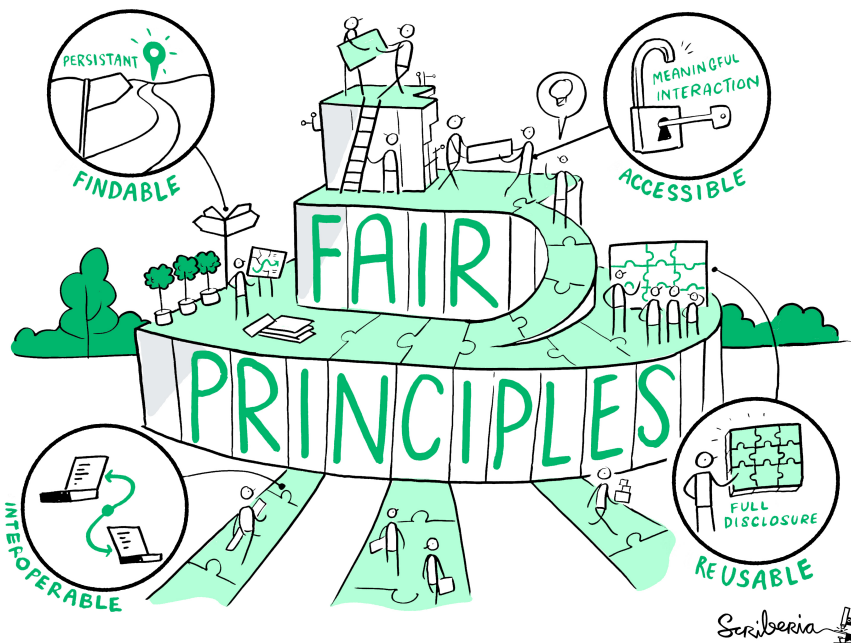
- **Handling Inbound Copyleft (Scenario 4 & Scenario 5):** When you copy non-trivial copyleft code snippets (e.g., CC BY-SA from Stack Overflow or GPL snippets) or link directly against a GPL library, your overall project becomes a combined work. Selecting a compatible copyleft license upfront (`GPL-3.0` or `EUPL-1.2`) satisfies the reciprocal sharing terms and allows the pipeline scanner to pass without conflict.
- **Handling Inbound Copyleft (Scenario 4 & Scenario 5):** Copying a non-trivial copyleft snippet (e.g., CC BY-SA code from Stack Overflow, or a GPL fragment) creates a combined work. Linking against a copyleft library may do the same, depending on the license and the linking method. In both cases, selecting a compatible copyleft license upfront (`GPL-3.0` or `EUPL-1.2`) satisfies the reciprocal terms and lets the scanner pass — and checking the exact SPDX identifier first avoids the `GPL-2.0-only` incompatibility trap.
- **Packaging and Build Automation (Scenario 6 & Scenario 7):** Keep plain-text build recipes (Dockerfiles) permissively licensed for maximum reuse, while annotating compiled container image binaries as multi-license aggregate bundles to satisfy embedded base-layer obligations.
- **AI Assets and Dual-Licensing (Scenario 8 & Scenario 9):** Run code-similarity scanners to catch LLM training memorization before releasing AI-assisted code, and apply dual-licensing to separate executable software code (`MIT`) from non-code datasets and model weights (`CC-BY-4.0`).
- **Standardized Distribution:** Adding machine-readable **SPDX headers** across every script, Dockerfile, and prompt template lets `reuse lint` confirm that every asset has a declared, documented license. Note what this does and does not prove: the linter verifies that declarations exist and are well-formed, not that they are legally correct or mutually compatible. Automation makes your intent auditable — it does not replace the judgment calls in the scenarios above.

Software citation

? Questions

- Is putting software on GitHub/GitLab/... publishing?
- Where to publish software?
- How can software be cited?
- How can I make my software citeable?

Is putting software on GitHub/GitLab/... publishing?



FAIR principles. (c) [Scriberia](#) for [The Turing Way](#), CC-BY.

Is it enough to make the code public for the code to remain **findable and accessible**?

- No. Because nothing prevents me from deleting my GitHub repository or rewriting the Git history and we have no guarantee that GitHub will still be around in 10 years.
- **Make your code citable and persistent:** Get a persistent identifier (PID) such as DOI in addition to sharing the code publicly, by using services like [Zenodo](#) or similar services.

How to make your software citable

Discussion (Citation-1): Explain how you currently cite software

- Do you cite software that you use? How?
- If I wanted to cite your code/scripts, what would I need to do?

Checklist for making a release of your software citable:

- ✓ Assigned an appropriate license
- ✓ Described the software using an appropriate metadata format
- ✓ Clear version number
- ✓ Authors credited
- ✓ Procured a persistent identifier
- ✓ Added a recommended citation to the software documentation

This checklist is adapted from: N. P. Chue Hong, A. Allen, A. Gonzalez-Beltran, et al., Software Citation Checklist for Developers (Version 0.9.0). Zenodo. 2019b. ([DOI](#))

Our practical recommendations:

- Add a file called [CITATION.cff](#) (example).
- Get a [digital object identifier \(DOI\)](#) for your code on [Zenodo](#) (example).
- [Step-by-step recipe](#) for how to make your GitHub project citable using [Zenodo](#).
- Make it as easy as possible: clearly say what you want cited.

This is an example of a simple `CITATION.cff` file:

```
cff-version: 1.2.0
message: "If you use this software, please cite it as below."
authors:
  - family-names: Doe
    given-names: Jane
    orcid: https://orcid.org/1234-5678-9101-1121
title: "My Research Software"
version: 2.0.4
doi: 10.5281/zenodo.1234
date-released: 2021-08-11
```

More about `CITATION.cff` files:

- [GitHub now supports CITATION.cff files](#)
- [Web form to create, edit, and validate CITATION.cff files](#)
- [Video: "How to create a CITATION.cff using cffinit"](#)

Papers with focus on scientific software

Where can I publish papers which are primarily focused on my scientific software? Great list/summary is provided in this blog post: ["In which journals should I publish my software?"](#) (Neil P. Chue Hong)

How to cite software

Great resources

- A. M. Smith, D. S. Katz, K. E. Niemeyer, and FORCE11 Software Citation Working Group, "Software citation principles," PeerJ Comput. Sci., vol. 2, no. e86, 2016 ([DOI](#))
- D. S. Katz, N. P. Chue Hong, T. Clark, et al., Recognizing the value of software: a software citation guide [version 2; peer review: 2 approved]. F1000Research 2021, 9:1257 ([DOI](#))
- N. P. Chue Hong, A. Allen, A. Gonzalez-Beltran, et al., Software Citation Checklist for Authors (Version 0.9.0). Zenodo. 2019a. ([DOI](#))

- N. P. Chue Hong, A. Allen, A. Gonzalez-Beltran, et al., Software Citation Checklist for Developers (Version 0.9.0). Zenodo. 2019b. ([DOI](#))

Recommended format for software citation is to ensure the following information is provided as part of the reference (from [Katz, Chue Hong, Clark, 2021](#) which also contains software citation examples):

- Creator
- Title
- Publication venue
- Date
- Identifier
- Version
- Type

! FAIR for Research Software (FAIR4RS)

The FAIR4RS Principles adapt the original FAIR data principles to improve the **Findability, Accessibility, Interoperability, and Reusability** of research software. They provide guidelines to make research software more open, discoverable, citable, and reusable, which supports reproducibility and open science. The summary of the FAIR4RS below is a compressed version of the principles listed in [Barker, Michelle, et al. "Introducing the FAIR Principles for research software." Scientific Data 9.1 \(2022\): 622](#).

- **Findable:** Software should have a globally unique and persistent identifier (e.g. clear versioning of releases, DOI from Zenodo when depositing a software release, software metadata in citation file).
- **Accessible:** Software and metadata should be retrievable using open protocols (e.g. downloading from GitHub or Zenodo).
- **Interoperable:** Software should use open formats and standards to work with other tools (e.g. input/output files in CSV, dependencies listed in `requirements.txt` or `environment.yml`, configuration files in yaml).
- **Reusable:** Software should have clear licensing, documentation, and provenance (e.g. a standard license and a `README` with usage instructions, authors listed with ORCIDs).

There are great resources to self-evaluate the *FAIRness* of your research software:

- [FAIR Software Checklist](#) which provides a questionnaire and even a badge of how FAIR the software is

- [FAIR Software NL](#) highlights with nice visuals the five most important elements of FAIR4RS (1. Public accessible repository with version control; 2. License; 3. Software registry; 4. Software citation file; 5. Software quality checklist)

Sharing data

Services for sharing and collaborating on research data

To find a research data repository for your data, you can search on the [Registry of Research Data Repositories \(re3data\)](#) platform and filter by country, content type, discipline, etc.

International:

- [Zenodo](#): A general-purpose open access repository created by OpenAIRE and CERN. Integration with GitHub, allows researchers to upload files up to 50 GB.
- [Figshare](#): Online digital repository where researchers can preserve and share their research outputs (figures, datasets, images and videos). Users can make all of their research outputs available in a citable, shareable and discoverable manner.
- [EUDAT](#): European platform for researchers and practitioners from any research discipline to preserve, find, access, and process data in a trusted environment.
- [Dryad](#): A general-purpose home for a wide diversity of datatypes, governed by a nonprofit membership organization. A curated resource that makes the data underlying scientific publications discoverable, freely reusable, and citable.
- [The Open Science Framework](#): Gives free accounts for collaboration around files and other research artifacts. Each account can have up to 5 GB of files without any problem, and it remains private until you make it public.

Sweden:

- [Researchdata.se - research data from a wide range of disciplines](#)
- [ICOS for climate data](#)
- [Bolin center climate / geodata](#)
- [NBIS for life science, sequence, omics data](#)

Norway:

- [NIRD archive is widely used by some communities](#)
- [NSD - Norwegian Center for Research Data, for any kind of data](#)
- [Dataverse.no - Dataverse network, based at University of Tromsø but open for other institutions](#)
- [ELIXIR Norway for life science, sequence, omics data](#)

Denmark:

- [Datavejviser](#) – metadata catalogue aggregating from research organisations and National Archives
- [GEUS Dataverse](#) – data from the Geological Survey of Denmark and Greenland
- [DTU Data](#) – repository of the Technical University of Denmark
- [DeiC Dataverse for University of Copenhagen](#)

Finland:

- [IDA, for general data](#)
- [FSD for social data](#)
- [more information on Finnish resources for open data](#)

Portugal:

- [DMPortal - Dataverse network for any research data](#)

Resources for data management

- [DataLad](#)
- [Data Stewardship Wizard](#)

Licensing of datasets and databases

- The EU has a [database directive](#) which restricts data mining on databases.
- Has a somewhat similar effect to copyright, because copyright would not apply to data mining.
- A good license also gives rights to data mine. So not a major concern.

When you can use datasets:

- The license allows
- Your country has exceptions for research
- The data doesn't come from the EU

License text, slides, images, and supporting information under a [Creative Commons license](#), and get a DOI using [Zenodo](#) or [Figshare](#) or [OSF](#) other services.

Licensing and machine learning/ AI

This section is maybe more relevant to **developers of AI models / AI systems** rather than **users of AI models / AI systems**.

Is it data? Is it software? It depends. We need to consider the AI system as a whole, the training data, the production data, the AI output, and how it is put on service. **AI models** are like the engine of the car: they cannot do anything without the rest of the car infrastructure. **AI systems** are the whole car with the AI model and all the software and hardware to actually use it.

Depending on what you are going to share, there might be things to consider beyond the license.

For example **large language models** are often shared with open source software licenses, on **HuggingFace** which is like a GitHub/GitLab for AI models (see for example the [OLMO model](#)[↗]). Many so-called *open-source* models are actually just *open-weights* models: only the trained neural network weights are shared, while the training data, training code, and full documentation are often kept private. This lack of transparency raises concerns about reproducibility and accountability and this phenomenon is sometimes called “**open washing**” ([ref](#)[↗]). Models are also shared with a **model card** which is a documentation tool for transparency that provide a comprehensive snapshot of a model’s characteristics and ethical considerations (see [Ch.8 Glerean 2025](#)[↗]).

What about ethics? What about liability?

As AI models (e.g. the deep network weights) and AI systems (the model with all the software and infrastructure to query it) are becoming more available, there can be legal (and ethical!) requirements on the developer of the AI model/system by the [EU AI Act](#)[↗]. In general researchers do not need to worry, but ethically one should consider that if the research-purpose AI model/system could be used for something harmful, ethically (if not legally) one should consider if such model/system should be implemented at all.

What about the training data inside the model? Large models can memorize and unintentionally reveal parts of their training data. This raises concerns about copyright, trade secrets, and personal data. News publishers and artists are suing AI companies for unauthorized use of their content in training. It is still unclear how traditional data licenses can apply to data that has been transformed into model weights.

More resources

- [RAIL initiative: “Responsible AI licenses”](#)[↗]
- [The Turing Way: Machine Learning Model Licenses](#)[↗]

- [“Expert Q&A on Artificial Intelligence \(AI\) Licensing”](#) 
-

Further reading

- [The Turing way](#) 
- [Illustrations from the Turing Way book dashes](#) 
- [Reproducibility syllabus](#) 
- The [reproducible research data analysis platform](#) 
- Good talks on open reproducible research can be found [here](#) 
- [“FAIR is not fair enough”](#) 
- [“A FAIRer future”](#) 
- [“Top 10 FAIR Data & Software Things”](#)  are brief guides that can be used by the research community to understand how they can make their research (data and software) more FAIR.
- [Five recommendations for fair software](#) 
- [Publishing research software](#)  A MIT libraries webpage on why to publish software, where to publish software, and how to make software citable.
- [Software Quality Checklist](#) 
- [MolSSI Best Practice Guides](#) 
- [Five recommendations for fair software](#) 
- [Awesome Research Software Registries](#) 

Who is the course for?

- Someone writing their own relatively small material
- Someone who wants to incorporate other code/libraries into their own projects
- Group leader who wants to decide how to balance openness and long-term strategy

List of exercises

Full list

This is a list of all exercises and solutions in this lesson, mainly as a reference for helpers and instructors. This list is automatically generated from all of the other pages in the lesson. Any single teaching event will probably cover only a subset of these, depending on their interests.

Instructor guide

Basics

This is basically presented as-is - not much technical presentation is needed.

There are the following main acts:

- First section is about social coding and sharing code and motivation for sharing code with a discussion that licensing alone isn't enough. If you want people to use your work, you need to do more to make your work easily reusable.
- Emphasis of the concept of derivative work and that everything is based on other things. And if your goal is citations, you want as many people using your work as possible, because then they can cite you.
- Overview of license types. There are many online resources, and the biggest question is permissive vs. weak copyleft vs. strong copyleft. The [cake analogy for licenses](#) [↗] is only linked and can be skipped and does not have to be discussed in detail.
- Then a brief section on code citations. It's good if you manage to start a discussion around this topic.
- The "Sharing data" part is meant to bring awareness about FAIRness and Open Science and to show how DOIs can be obtained for Git repositories.

Hints

Before you begin, it's worth thinking about some particular good or bad cases you know about from your experience.

- When have you lost access or not been able to use something because licensing wasn't handled well.
- When did multiple ownership mess up your plans?

This talk used to be called "software licensing". You could mention that to make it more interesting, we have adopted the current broader name, and our emphasis is not just licensing, but why licensing is needed.

OLD Software licensing

! Objectives

- Knowing about what derivative work is and whether we can share it.
- Get familiar with terminology around licensing.
- Practical advice for software licensing.

Copyright



- **Trademark:** Protects a name/brand from impersonation.
- **Patent:** Protects a novel, non-obvious, technical invention.
- **Copyright:** Protects **creative expression**: software, writing, graphics, photos, certain datasets, this presentation. Practically “forever” (lifetime of author + 70 years).

Copyright controls whether and how we can distribute the original work or the **derivative work**.

Derivative work: Sampling/remixing



[Midjourney, CC-BY-NC 4.0]



[Midjourney, CC-BY-NC 4.0]

- Changing and distributing software is similar to changing and distributing music
- You can do almost anything if you don't distribute it

Often we don't have the choice:

- We are expected to publish software
- Sharing can be good insurance against being locked out

Exercise: Derivative work

Licensing-1: What constitutes derivative work?

This question 5 below can be used as a starting point and copied to the collaborative document or form input for an online poll:

Question 5: Which of these are derivative works?

****Choose many****. Vote by adding an `o` character:

- A. Download some code from a website and add on to it
- votes:
- B. Download some code and use one of the functions in your code
- votes:
- C. Changing code you got from somewhere
- votes:
- D. Extending code you got from somewhere
- votes:
- E. Completely rewriting code you got from somewhere
- votes:
- F. Rewriting code to a different programming language
- votes:
- G. Linking to libraries (static or dynamic), plug-ins, and drivers
- votes:
- H. Clean room design (somebody explains you the code but you have never seen it)
- votes:
- I. You read a paper, understand algorithm, write own code
- votes:

✓ Solution

- Derivative work: A-F
- Not derivative work: G-I
- E and F: This depends on how you do it, see clean room design.

📌 Plagiarism vs. Intellectual Property Rights = Research Ethics vs Law

This insert can be skipped and left as reading exercise

In academic context it is important to consider also *plagiarism* and how it relates to copyright and more broadly Intellectual Property Rights ([a clear explanation at this page](#)[🔗]). Plagiarism is the practice of taking somebody else's ideas or work and claim them as your own: it is the **unacknowledged** use of another person's work. Intellectual Property Rights (IPRs) infringement instead is the **unauthorised** use of another's work.

IPRs can be classified in two main groups ([WTO](#)[🔗]): i) Copyright and rights related to copyright (computer programs are here) and ii) Industrial properties like trademarks, and inventions (which may include specific technical implementations of systems or code) protected by patents.

In research ethics, plagiarism is one of the three definition of research misconduct (along with *fabrication* and *falsification*, see ALLEA, [European Code of Conduct for Research Integrity](#)[↗]). Plagiarism is not illegal per se, but it can lead to serious consequences like the retraction of published work. One can engage in plagiarism, without necessarily breaking any IPR law (e.g. write a new book by reusing the plot of an old book that is not under copyright anymore). Copyright infringement instead is illegal and it can result in criminal charges (e.g. fines). Copyright however protects the particular expression of an idea or fact (for example, the specific source code of a program, but not the underlying algorithm itself).

There is no pre-defined “number of lines of code”, “seconds of a song”, or “pixels of an image” that can clearly set the basis for plagiarism or IPR infringement. However in the context of research, it can be possible to use *Quotation Exception* (in EU, [ref](#)[↗]) and *Fair use* (in USA, [ref](#)[↗]). Fair use has become controversial recently as it is used as legal basis for training large language models based on scraped internet data (See for example Henderson, P., Li, X., Jurafsky, D., Hashimoto, T., Lemley, M. A., & Liang, P. (2023). [Foundation models and fair use. Journal of Machine Learning Research, 24\(400\), 1-79.](#)[↗])

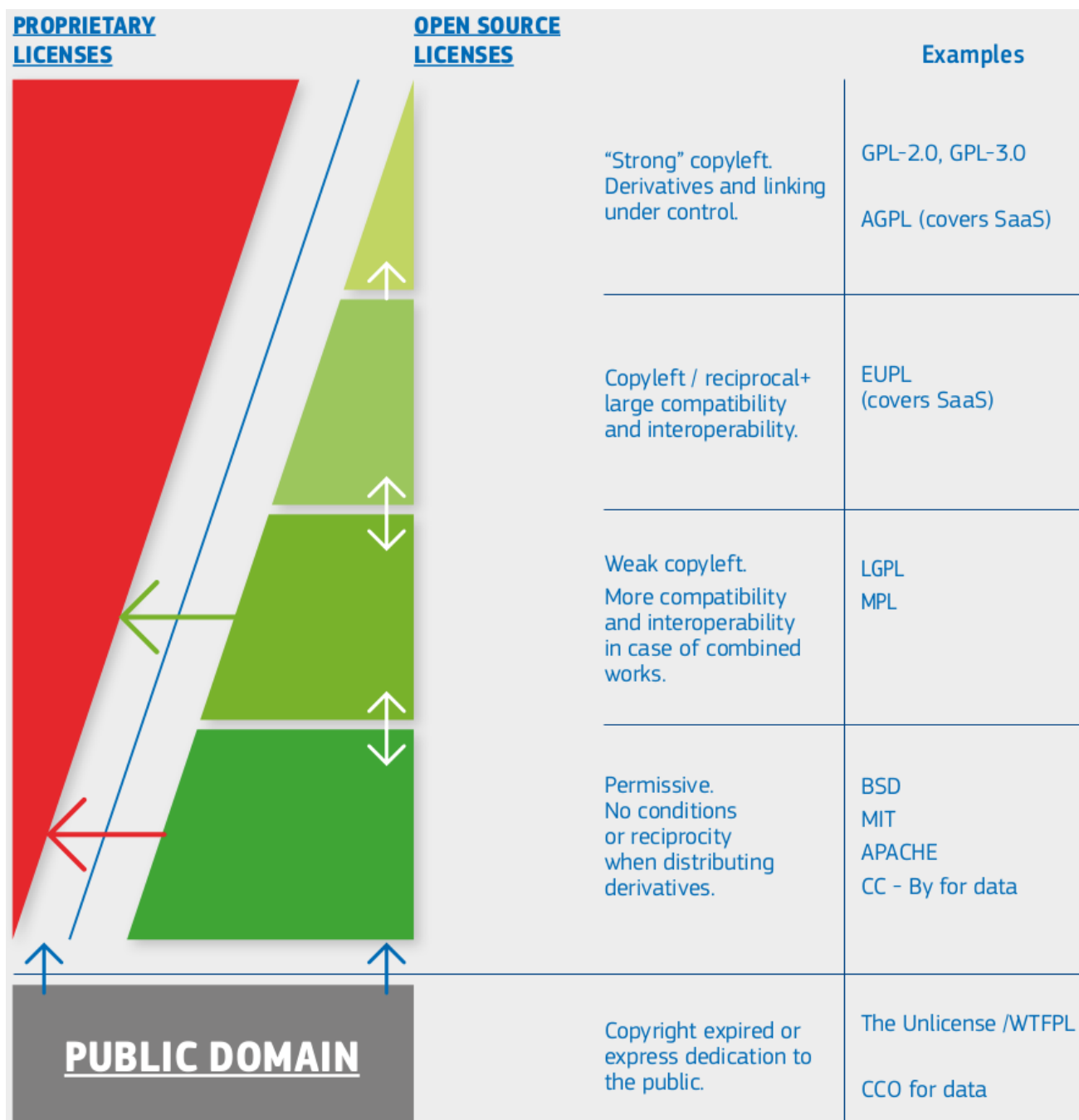
Derivative work and containers

Containers are a bit more tricky when it comes to licenses.

- Distribution of container recipes: it's like distributing source code
- Distribution of container images: it can be considered like distributing a binary compiled software

The latter case is a bit more nuanced and the interested reader should read more about “Mere Aggregation” at [GPL-FAQ](#)[↗]. Briefly, if the container image just bundles separate programs that talk through normal system interfaces, it is an **aggregate** and each keeps its own license (like a CD-ROM with various packages). If the components are tightly integrated into one program (e.g. a pipeline with various parts that the container can run as a single program), the image may be treated as a **derivative work**, and stricter license obligations (e.g. GPL copyleft) can apply.

Taxonomy of software licenses



European Commission, Directorate-General for Informatics, Schmitz, P., European Union Public Licence (EUPL): guidelines July 2021, Publications Office, 2021,

<https://data.europa.eu/doi/10.2799/77160>

Comments:

- Arrows represent compatibility (A -> B: B can reuse A)
- Proprietary/custom: Derivative work typically not possible (no arrow goes from proprietary to open)
- Permissive: Derivative work does not have to be shared
- Copyleft/reciprocal: Derivative work must be made available under the same license terms
- NC (non-commercial) and ND (non-derivative) exist for data licenses but not really for software licenses

Great resource for comparing software licenses: [Joinup Licensing Assistant](#)

- Provides comments on licenses
- Easy to compare licenses ([example](#))
- [Joinup Licensing Assistant - Compatibility Checker](#)
- Not biased by some company agenda

If you would like to learn more about licenses, check out our slide deck: [“Software licensing and open source explained with cakes”](#).

Exercise: Licensing situations

Licensing-2: Consider some common licensing situations

1. What is the StackOverflow license for code you copy and paste?
2. A journal requests that you release your software during publication. You have copied a portion of the code from another package, which you have forgotten. Can you satisfy the journal's request?
3. You want to fix a bug in a project someone else has released, but there is no license. What risks are there?
4. How would you ask someone to add a license?
5. You incorporate MIT, GPL, and BSD3 licensed code into your project. What possible licenses can you pick for your project?
6. You do the same as above but add in another license that looks strong copyleft. What possible licenses can you use now?
7. Do licenses apply if you don't distribute your code? Why or why not?
8. Which licenses are most/least attractive for companies with proprietary software?

✓ Solution

1. As indicated [here](#), all publicly accessible user contributions are licensed under [Creative Commons Attribution-ShareAlike](#) license. See Stackoverflow [Terms of service](#) for more detailed information.
2. “Standard” licensing rules apply. So in this case, you would need to remove the portion of code you have copied from another package before being able to release your software.
3. By default you are not authorized to use the content of a repository when there is no license. And derivative work is also not possible by default. Other risks: it may not be clear whether you can use and distribute (publish) the bugfixed code. For the repo owners it may not be clear whether they can use and distributed the bugfixed code. However, the authors may have forgotten to add a license so we suggest you to contact the authors (e.g. make an issue) and ask whether they are willing to add a license.
4. As mentioned in 3., the easiest is to fill an issue and explain the reasons why you would like to use this software (or update it).

5. Combining software with different licenses can be tricky and it is important to understand compatibilities (or lack of compatibilities) of the various licenses. GPL license is the most protective (BSD and MIT are quite permissive) so for the resulting combined software you could use a GPL license. However, re-licensing may not be necessary.
6. Derivative work would need to be shared under this strong copyleft license (e.g. AGPL or GPL), unless the components are only plugins or libraries.
7. If you keep your code for yourself, you may think you do not need a license. However, remember that in most companies/universities, your employer is “owning” your work and when you leave you may not be allowed to “distribute your code to your future self”. So the best is always to add a license!
8. The least attractive licenses for companies with proprietary software are licenses where you would need to keep an open license when creating derivative work. For instance GPL and AGPL. The most attractive licenses are permissive licenses where they can reuse, modify and relicense with no conditions. For instance MIT, BSD and Apache License.

When should I add a license?

Choose a license early in the project, even before you publish it. Later in the project it may become complicated to change it. Agreeing on a software license does not mean that you have to make it open immediately. You can also follow the **“open core” approach**: You don’t have to open source all your work. Core can be open and on a public branch. Unpublished code can be on a private repository.

However, we recommend to **work as if the code is public even though it still may be private** (thanks to E. Glerean for this great suggestion): This is to avoid surprises about code in the history with incompatible license years later when you decide to open the project.

How to add a license if your work is derivative work

Your code is derivative work if you have started from an existing code and made changes to it or if you incorporated an existing code into your code.

If your code is derivative work, then **you need to check the license of the original code**. Depending on the license, your choices might be limited. In this case we recommend to use these two resources:

- [Joinup Licensing Assistant - Find and compare software licenses](#) 
- [Joinup Licensing Assistant - Compatibility Checker](#) 

If the original code does not have a license, you may not be able to distribute your derivative code. You can try to contact the authors and ask them to clarify the license of their code.

Practical steps for **incorporating something small into your own project** with a license that allows you to do so (as an example incorporating a function or two from another project):

- Create a `LICENSES/` folder in your project and “put the unmodified license text (i.e., the license text template without any copyright notices) in your `LICENSES/` folder” (<https://reuse.software/faq/#license-templates>). This way if you reuse code from multiple projects, you can keep there multiple license files.
- **Put the code that you incorporate into a separate file or separate files.** This makes it later easier to see what was incorporated, and what was written from scratch. On top of the file(s) which you have incorporated into your project add (and adapt) the following header ([more examples](#)):

```
# SPDX-FileCopyrightText: 2023 Jane Doe <jane@example.com>
#
# SPDX-License-Identifier: MIT
```

The [REUSE](#) initiative was started by the [Free Software Foundation Europe](#) to make licensing of software projects easier. It is OK if you prefer to not follow this strict format but the advantage of following it is that the [reuse-tool](#) makes it then easy to verify and update license headers if you have many files from different sources.

- If it does not make sense to have several files in your project (e.g. when incorporating something into a notebook), then add a note/comment about the license and where the code came from on top of the function.
- Although it is not dictated by the license but it can still be nice to acknowledge the incorporated functions/code in your README/documentation and to cite their work if you publish a paper about your code.
- Some licenses are more permissive (you can keep your changes private) but some licenses require you to publish the changes (share-alike).

Practical steps for making **changes to an existing project** with a license that allows you to do so:

- If the project is on GitHub or GitLab or similar, first fork the project (copy it into your user space where you can make changes).
- For the BSD and MIT licenses you are not obliged to state your changes but it can still be helpful for others if you do. You can state your changes in the header of the files you have modified. It can be helpful to state bigger-picture changes in the README file of the project.
- Some licenses are more permissive (you can keep your changes private) but some licenses require you to publish the changes (share-alike).

If your work is not derivative work

If you have started “from scratch”, and not used any existing code, or incorporated existing code into your code, then you may consider your code to be not derivative work.

Before you may choose a license, clarify the following points with, for example, your supervisor, collaborators, or principal investigator:

- Does your work contract, grant, or collaboration agreement dictate a specific license?
- Is there an intent to commercialize the code?
- When there is unknown or mixed ownership: If there are multiple persons or organizations as owners of the code, all must agree to the license.

Do not invent your own license. Choose one of the standard licenses, otherwise compatibility is not clear:

- [Joinup Licensing Assistant - Find and compare software licenses](#)
- [Joinup Licensing Assistant - Compatibility Checker](#)

Practical steps:

- Create a `LICENSES/` folder ([example](#)).
- Put the unmodified license text (i.e., the license text template without any copyright notices) in plain text format into the folder ([example](#)). Here are the two above licenses in plain text: [EURL](#) and [MIT](#) (but the latter contains a copyright notice which we rather want to have on top of files).
- Add copyright and license information to each file following <https://reuse.software/tutorial/> which uses a standard format with so-called [SPDX identifiers](#). Example below ([example](#)):

```
# SPDX-FileCopyrightText: 2023 Jane Doe <jane@example.com>
#
# SPDX-License-Identifier: EUPL-1.2
```

The [REUSE](#) initiative was started by the [Free Software Foundation Europe](#) to make licensing of software projects easier. It is OK if you prefer to not follow this strict format but the advantage of following it is that the [reuse-tool](#) makes it then easy to verify and update license headers if you have many files from different sources.

- For really small projects with one or two files the above may seem excessive and some projects choose to not have copyright information on top of their files and they only have one `LICENSE` file and that is OK for really small projects.

With generative AI tools for coding such as GitHub copilot, Cursor, or even basic chat implementations (ChatGPT, Claude, Grok, ...) the responsibility fully lays on the person who is going to use (and publish) the generated code. You can never blame the autopilot or the company who invented it, only the driver (you!).

There are various risks in using generative AI code (this is not a taxonomy). A few examples:

- Risks for the derivative work: you think your code is doing what you asked, but you did not review it and your results are false
- Risks for the system in use: your generated code has software security issues, e.g. an import is a *typosquat* of an actual library (e.g. “microsoft” is spelled “rnicrosoft” and depending on the font you might totally miss it...)
- Risks related to licenses/IPR: you have generated code that is actually verbatim copy of fully copyrighted code, or code that requires a strict copyleft license. Plagiarism (ethics) also applies.

If we focus on the last one, a recent paper ([ref](#)) estimates that around 2% of AI generated code is “strikingly similar to existing open-source implementations”. Generative AI tools are typically not able to provide an exact reference of where certain bits of generated code were copied from, so it is the responsibility of the researcher to verify that the produced code is citing and referencing the license of other published pieces of software. Possibly, future AI systems for code generation can be trained on code that share the same set of licenses (e.g. based only on MIT) to mitigate these risks.

Great resources

- [Research institution policies to support research software \(compiled by the Research Software Alliance\)](#)
- Guide from the Aalto University in Finland: “[Opening your Software at Aalto University](#)”
- [Draft: Research software licensing guide](#)
- [Joinup Licensing Assistant - Find and compare software licenses](#)
- [Joinup Licensing Assistant - Compatibility Checker](#)
- [Social coding lesson material](#) by [CodeRefinery](#)
- [Citation File Format \(CFF\)](#)
- [License Selector](#)
- [GitHub licensing guide](#)
- [Choosing an open-source licence](#)
- [Understanding Open Source and Free Software Licensing](#)
- [Software Licenses in Plain English](#)
- [Don's Bibliography of Ethical Source Reading and Resources](#)

- [Mikko Välimäki: The Rise of Open Source Licensing](#)
- [Lawrence Rosen: Open Source Licensing](#)
- [Aalto IPR Cheatsheet](#)
- [Contributor License Agreements](#)
- [4OSS recommendations](#)
- [4OSS lesson](#)
- [Intellectual Property Rights \(IPR\), Licensing And Patents](#)
- [Dispelling Open Source Confusion: An Introduction to Licenses](#)
- <https://choosealicense.com> (can send automatic pull request to your GitHub repo)
- <https://hintjens.gitbooks.io/social-architecture/content/chapter2.html>
- <http://rkd.zgib.net/scicomp/open-science/open-science.html>
- Nadia Asparouhova (formerly Nadia Eghbal): “Working in Public: The Making and Maintenance of Open Source Software” (Stripe Press)
- [Open Source Guides](#)
- [The Architecture of Open Source Applications](#)
- Christopher M. Kelty: “Two Bits: The Cultural Significance of Free Software” (Duke University Press, 2008)
- [Open Source \(Almost\) Everything](#)
- [99 ways to ruin an open source project](#)
- [Open Source Casebook](#)

Keypoints

- **You cannot ignore licensing:** default is “no one can make copies or derivative works”.

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